

# Kashif Riyaz

*AI Engineer & Backend Developer*

|                       |                                                    |
|-----------------------|----------------------------------------------------|
| <b>Date of Birth</b>  | 11.01.2001                                         |
| <b>Place of Birth</b> | Sopore, India                                      |
| <b>Address</b>        | Heiner-Fleischmann-Straße 1, 92224 Amberg, Germany |
| <b>Phone</b>          | (+49) 01590 2287258                                |
| <b>Email</b>          | kashifriyaz03@gmail.com                            |
| <b>LinkedIn</b>       | linkedin.com/in/kashif-riyaz-04170b16a             |
| <b>GitHub</b>         | github.com/kashif003                               |

## Work Experience

---

|                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 03/2025 – Present | <b>Student Assistant (Backend Developer)</b><br><i>Justus Liebig University, Giessen, Germany</i> <ul style="list-style-type: none"><li>Responsible for migration of analytical software tools from R to Python to improve performance and maintainability.</li><li>Developed automatic track detection systems in HPTLC utilizing semantic segmentation techniques.</li><li>Implemented unsupervised learning algorithms for automated chemical material detection in HPTLC images.</li></ul> |
| 05/2022 – 08/2022 | <b>Student Intern</b><br><i>MARS Exploration Pvt Ltd, Remote</i> <ul style="list-style-type: none"><li>Conducted Finite Element Analysis (FEA) of mechanical components using ANSYS.</li><li>Designed and optimized 3D parts and assemblies using SolidWorks.</li><li>Prepared technical documentation and presented key simulation results.</li></ul>                                                                                                                                         |
| 09/2022 – 10/2022 | <b>Student Intern</b><br><i>Indwell Automation, Mangalore, India</i> <ul style="list-style-type: none"><li>Configured and programmed PLCs for basic industrial automation tasks.</li><li>Designed HMI/SCADA screens to monitor and control plant processes in real-time.</li></ul>                                                                                                                                                                                                             |

## Academic Projects

---

|                   |                                                                                                                                                                                                         |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 11/2025 – Present | <b>Model Pruning &amp; Optimization</b> <ul style="list-style-type: none"><li>Implementing structured and unstructured pruning techniques to compress transformers while preserving accuracy.</li></ul> |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                   |                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                   | <ul style="list-style-type: none"> <li>Analyzing the trade-off between model sparsity and inference latency for deployment on resource-constrained devices.</li> <li>Fine-tuning pruned models to recover accuracy loss post-compression.</li> </ul>                                                                                                                                                    |
| 05/2025 – 09/2025 | <b>Autonomous Mobile Robot Simulation (Webots)</b> <ul style="list-style-type: none"> <li>Developed a ROSbot simulation in Webots to enable autonomous navigation capabilities.</li> <li>Implemented SLAM (Simultaneous Localization and Mapping) for real-time environment mapping.</li> <li>Engineered path planning and path following algorithms for collision-free autonomous movement.</li> </ul> |
| 03/2025 – 06/2025 | <b>Predictive Fault Detection in Wind Turbines</b> <ul style="list-style-type: none"> <li>Built clustering models for early fault detection in wind turbines using sensor data.</li> <li>Performed data preprocessing, feature extraction, and anomaly detection.</li> </ul>                                                                                                                            |
| 09/2024 – 10/2024 | <b>Robotic Arm Mastering Tic-Tac-Toe</b> <ul style="list-style-type: none"> <li>Developed and implemented an autonomous Tic-Tac-Toe playing robotic arm.</li> <li>Enhanced human-robot interaction through innovative gaming experiences.</li> </ul>                                                                                                                                                    |
| 03/2024 – 07/2024 | <b>Explainable AI Using Grad-CAM in PyTorch</b> <ul style="list-style-type: none"> <li>Investigated the underlying behavior of (deep) neural networks to explain model performance.</li> <li>Visualized gradient-weighted class activation mappings (Grad-CAM).</li> </ul>                                                                                                                              |
| 01/2024 – 03/2024 | <b>Python Framework for Data Acquisition via Web Scraping</b> <ul style="list-style-type: none"> <li>Scraped news articles from websites to build a structured database.</li> <li>Preprocessed and developed machine learning models using the curated news dataset.</li> </ul>                                                                                                                         |

## Education

|                   |                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 03/2024 – Present | <b>Master of Artificial Intelligence</b><br><i>Ostbayerische Technische Hochschule (OTH), Amberg, Germany</i> <ul style="list-style-type: none"> <li>Focus: Industrial Applications.</li> <li>Focused on building machine learning and deep learning models.</li> <li>6+ months of hands-on experience in backend software development, model development, and evaluation.</li> </ul> |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Skills

|           |                                                                                                                     |
|-----------|---------------------------------------------------------------------------------------------------------------------|
| Languages | English: Proficient (C1 Listening / B2 Speaking & Writing) · German: Basic Knowledge (A2) · Kashmiri: Mother tongue |
| Software  | Python, PyTorch, TensorFlow, Docker, Git, R (migration experience)                                                  |

**Data Science**      Feature Engineering, Data Analysis, Deep Learning, Model Training & Optimization, Model Deployment (MLOps)

## Certificates & Awards

---

- Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning (Deeplearning.AI)
- Introduction to Machine Learning Course (Authorized by Google)
- Data Visualization (Authorized by Kaggle)